

**Indian Institute of Technology Bombay**

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*CE 749 : Freight Transportation Planning and Logistics*

*Course Project Report*

**Bi-Objective Optimization of Freight Allocation  
on the Indian Railway Network**

**(A Multi-Commodity Flow Approach with  
Transshipment, Backhaul, and Green Routing)**

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## Abstract

Indian Railways transported 1,509 MT of freight in FY 2022-23, with the Golden Quadrilateral corridor carrying 55% of this volume over just 16% of total route length. The recent commissioning of the Eastern and Western Dedicated Freight Corridors has introduced fully electrified, high-capacity alternatives alongside existing diesel conventional lines, creating a structural conflict between cost minimisation and CO<sub>2</sub> emission reduction in freight routing decisions.

This report formulates a **Bi-Objective Multi-Commodity Flow Linear Program (MCF-LP)** for annual freight allocation on the Indian Railways Golden Quadrilateral network. The model routes four commodity types : Coal, Cement, Foodgrains, and Fertilizers across a 16-node, 50-arc network representing the GQ and its diagonals, with a total annual demand of 435 MT across 28 origin-destination pairs. Three operationally realistic features distinguish the formulation from standard MCF models: (i) transshipment costs at all 12 intermediate junction nodes capturing marshalling yard detention charges; (ii) empty wagon backhaul repositioning costs modelled as a linear auxiliary objective component; and (iii) traction-differentiated emission factors (electric: 0.010 kgCO<sub>2</sub>/tonne-km; diesel: 0.035 kgCO<sub>2</sub>/tonne-km) sourced from the India Default GHG Emission Values report (SFC India, 2025). The two objectives : total logistics cost ( $Z_1$ ) and total CO<sub>2</sub> emissions ( $Z_2$ ) are optimised simultaneously using the **epsilon-constraint method**, which generates a 12-point Pareto frontier.

Results show that the emission-optimal solution achieves an **8.66% reduction in CO<sub>2</sub>** (from 10.22 Mt to 9.34 Mt) at a **cost premium of 15.45%** over the cost-optimal baseline of Rs 99,614 crore. The Pareto frontier reveals a steep structure near the emission-optimal end: approximately 9.6 percentage points of the cost penalty can be recovered by accepting just 1.6% less emission reduction. Empty wagon backhaul repositioning accounts for **25.8% of total logistics cost** a component absent from standard single-term MCF formulations. A Greedy Shortest Path heuristic benchmark shows that naive routing produces solutions that emit **9.69% more** CO<sub>2</sub> than the MCF-LP optimum, confirming the value of formal optimisation for environmental performance. Sensitivity analysis identifies diesel arc capacity of 70 MT as the binding constraint threshold, beyond which cost plateaus completely. The carbon price breakeven is estimated at Rs 178/tonne CO<sub>2</sub>, providing a concrete threshold for green routing policy on Indian Railways.

**Keywords:** Multi-commodity flow, bi-objective optimization, Indian Railways, epsilon-constraint, green freight, transshipment, backhaul, Pareto frontier, Golden Quadrilateral, CO<sub>2</sub> emissions.

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## 1 Introduction

Indian Railways is one of the largest freight rail networks in the world, transporting 1,509.10 million tonnes (MT) of goods in FY 2022-23 and generating freight revenue of Rs 1,60,158 crore [1]. With a route network spanning over 67,000 km and a fleet of more than 3 lakh wagons, it serves as the circulatory system of India's industrial economy, moving coal to power plants, cement to construction sites, foodgrains through the Food Corporation of India supply chain, and fertilizers to agricultural distribution hubs across the country.

Despite this scale, a critical planning challenge persists at the operational level: given a set of freight demands across multiple origin-destination (OD) pairs and multiple commodity types, *how should freight be allocated across the railway network so as to minimise total logistics cost?* This question is the freight allocation problem and it is not easy to solve. Every year, suboptimal routing decisions translate directly into unnecessary expenditure on longer hauls, avoidable marshalling yard detention charges, and inefficient empty wagon repositioning. Conversely, the growing policy emphasis on decarbonising freight transport introduces a second, competing objective: minimising CO<sub>2</sub> emissions. These two objectives are in structural conflict on the Indian Railways network, where lower-cost routes frequently pass through diesel-traction corridors that emit up to 3.5 times more CO<sub>2</sub> per tonne-kilometre than electrified alternatives [2].

The Golden Quadrilateral (GQ) and its two diagonals connecting Delhi, Mumbai, Chennai, and Kolkata form the backbone of Indian freight, carrying approximately 55% of total national freight traffic over just 16% of total route length [3]. Line capacity utilisation on key trunk routes of the GQ has consistently exceeded 115–150%, making it both the most critical and most congested part of the network. The recent commissioning of the Eastern Dedicated Freight Corridor (EDFC, 1,337 km) and the near-complete Western Dedicated Freight Corridor (WDFC, 1,506 km) has introduced fully electrified, high-capacity alternatives alongside existing diesel conventional lines. This creates a genuine and operationally significant cost-versus-emission tradeoff: the electric DFC routes are cleaner but carry higher per-tonne tariffs on longer alignments, while diesel corridors offer shorter routing at higher emission intensity. Determining the optimal allocation of freight between these competing corridors requires a formal bi-objective optimisation framework.

The academic literature on freight routing has addressed Multi-Commodity Flow (MCF) problems extensively (see [4] for a comprehensive review), and bi-objective green freight models have been studied in intermodal contexts [5]. However, several features that are critical to real-world Indian Railways planning are routinely absent from academic formulations. First, *transshipment costs* marshalling yard detention charges at intermediate junctions such as Nagpur, which is India's largest central freight junction are rarely modelled explicitly. Second, *empty wagon backhaul repositioning costs*, which arise

when directional demand imbalances require empty wagons to be returned against the primary flow direction, are omitted from most formulations despite accounting for a substantial fraction of total logistics cost. Third, most academic studies use synthetic or aggregated demand data rather than real zonal traffic statistics. These omissions limit both the realism and the policy relevance of existing models for the Indian context.

This report addresses these gaps by formulating and solving a **Bi-Objective Multi-Commodity Flow Linear Program (MCF-LP)** for annual freight allocation on the Indian Railways Golden Quadrilateral network. The model is applied to a 16-node, 50-arc network representing the GQ and its diagonals, routing four major commodity types Coal, Cement, Foodgrains, and Fertilizers across 28 OD pairs with a total annual demand of 435 MT. The formulation incorporates (i) real zone-wise OD demand derived from the IR Year Book 2022-23, (ii) transshipment costs at all 12 intermediate junction nodes, (iii) empty wagon backhaul repositioning costs modelled as a linear auxiliary, and (iv) differentiated emission factors for electric and diesel traction consistent with the India-specific GHG emission values published by Smart Freight Centre India [2]. The two objectives : total logistics cost and total CO<sub>2</sub> emissions are optimised simultaneously using the **epsilon-constraint method**, which generates a 12-point Pareto frontier characterising the complete cost-emission tradeoff.

Results show that the emission-optimal solution reduces CO<sub>2</sub> by **8.66%** at a cost premium of **15.45%** over the cost-optimal baseline of Rs 99,614 crore. Backhaul costs alone account for 25.8% of total logistics cost a finding with direct implications for IR's wagon utilisation policy. A Greedy Shortest Path heuristic benchmark demonstrates that naive routing produces solutions that emit 9.69% more than the MCF-LP optimum, confirming the practical value of formal optimisation for environmental performance. Sensitivity analysis identifies diesel arc capacity of 70 MT as the binding constraint threshold for the current demand profile.

The remainder of this report is organised as follows. Section 2 reviews the relevant literature on MCF, green freight, and railway routing models. Section 3 describes the network data and commodity demand. Section 4 presents the complete mathematical formulation. Section 5 describes the solution methodology including the epsilon-constraint algorithm and greedy heuristic. Section 6 presents and discusses all results. Section 7 concludes with policy implications and directions for future work.

## 2 Literature Review

The Multi-Commodity Network Flow (MCF) problem forms the methodological foundation of this work. This section reviews two directly relevant contributions that motivate the formulation and solution approach adopted here, followed by a summary of the research gap this work addresses.



## 2.1 Multi-Commodity Network Flow and Solution Methods

Malladi and Sowlati [4] present a systematic review of 263 MCF publications spanning 2000–2019, identifying freight transportation planning as the dominant application domain. Two findings are directly relevant here. First, the epsilon-constraint method is among the most widely adopted approaches for generating Pareto frontiers in bi-objective MCF problems, preferred over weighted-sum methods because it recovers non-dominated solutions even when the frontier is non-convex. Second, the review observes that real-world MCF applications consistently benefit from incorporating operational costs beyond arc traversal including terminal handling and repositioning yet most academic formulations omit these for tractability. This directly motivates the transshipment and backhaul extensions in the present model.

## 2.2 Bi-Objective Green Freight Optimisation

Demir et al. [5] develop a bi-objective model for intermodal freight transportation minimising cost and CO<sub>2</sub> simultaneously across a European network. Using the epsilon-constraint method, they find that nearly 6% of emissions can be eliminated at less than 1% additional cost, but the full emission-optimal solution requires a cost premium of approximately 10%. The present work finds a qualitatively similar Pareto structure on the Indian Railways network, with a steeper frontier (15.45% premium for 8.66% emission saving) reflecting the larger diesel-electric emission ratio ( $3.5\times$ ) specific to India's traction mix. Unlike Demir et al., the present model targets intra-rail traction mode choice rather than inter-modal mode shifts, and explicitly incorporates transshipment and backhaul costs that together account for 35.6% of total logistics cost.

## 2.3 Research Gap and Contribution

The two works above establish the MCF framework and epsilon-constraint method as appropriate for this problem, and confirm that operational cost components and emission differentiation are important model features. However, no published work applies these methods to Indian Railways specifically, nor simultaneously incorporates real zone-wise IR demand, junction transshipment costs, empty wagon backhaul, and traction-differentiated emission factors in a single bi-objective formulation. Table 2.1 positions this work relative to the reviewed literature.

**Table 2.1.** Comparison of This Work with Related Literature

| Feature                                | <i>Malladi &amp; Sowlati (2020)</i> | <i>Demir et al. (2019)</i> | <i>This Work</i> |
|----------------------------------------|-------------------------------------|----------------------------|------------------|
| Indian Railways context                | —                                   | —                          | ✓                |
| Real zone-wise demand                  | —                                   | —                          | ✓                |
| Bi-objective (cost + CO <sub>2</sub> ) | —                                   | ✓                          | ✓                |
| Epsilon-constraint Pareto              | —                                   | ✓                          | ✓                |
| Transshipment costs                    | —                                   | —                          | ✓                |
| Backhaul / empty wagon repositioning   | —                                   | —                          | ✓                |
| Traction-differentiated emissions      | —                                   | partial                    | ✓                |

### 3 Data and Network Description

#### 3.1 Network Structure

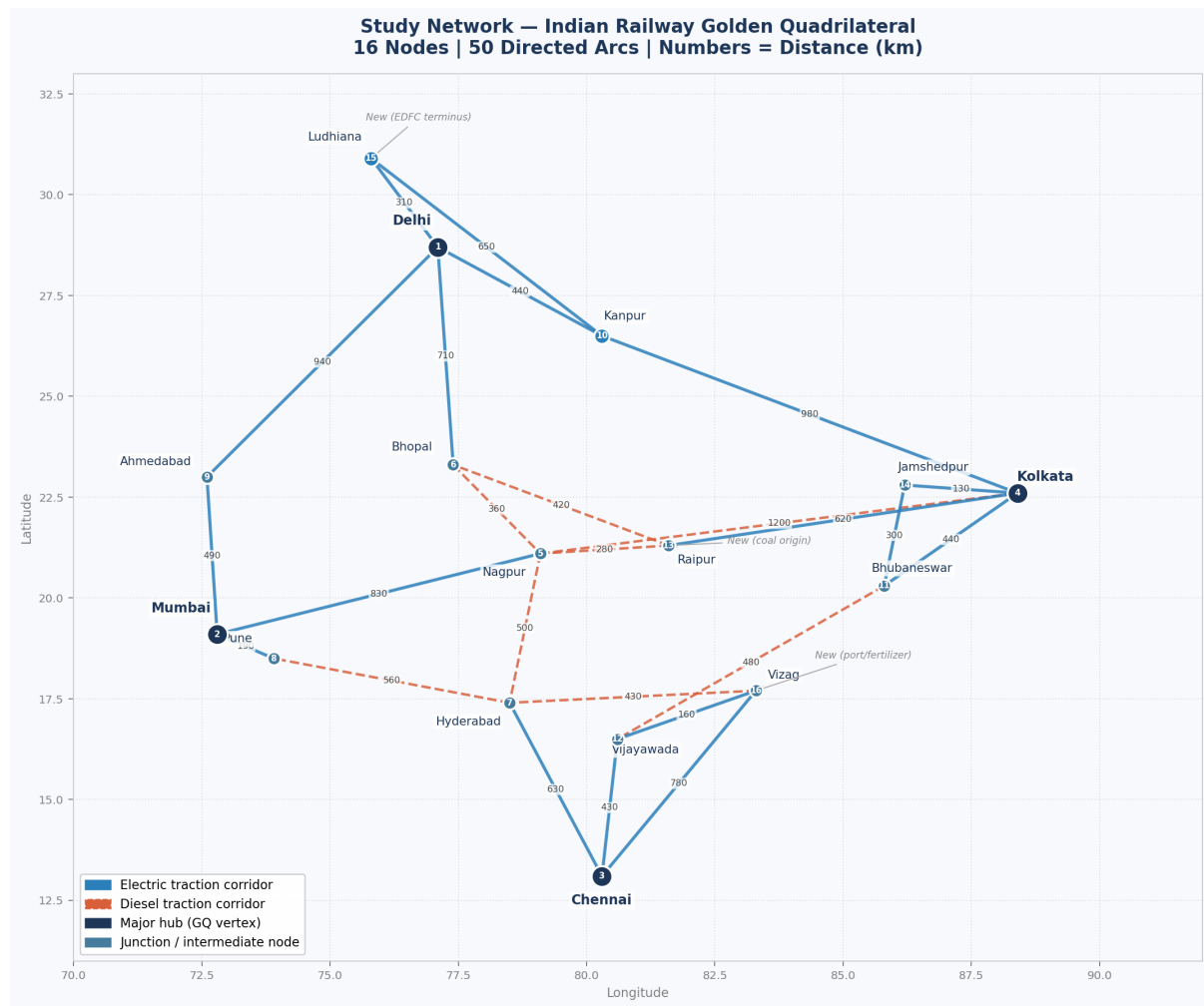
The study network represents the Indian Railways Golden Quadrilateral and its two diagonals, which together carry approximately 55% of total national rail freight over just 16% of total route length [3]. The network consists of **16 nodes**, **25 undirected corridors** (modelled as 50 directed arcs), and covers the major freight hubs, key junction cities, and industrial origin points of India. The four primary hubs Delhi, Mumbai, Chennai, and Kolkata form the vertices of the Golden Quadrilateral, while twelve additional nodes represent intermediate junctions, marshalling yards, and commodity origin/destination points.

Four nodes beyond the standard Golden Quadrilateral are included to improve the realism of conflict routing. Raipur (node 13) represents the South East Central Railway zone and is the gateway to SECL coal mines (Korba, Gevra, Kusmunda) the second largest coal producing cluster in India. Jamshedpur (node 14) anchors the eastern steel belt (Tata Steel, IISCO). Ludhiana (node 15) is the northern terminus of the Eastern Dedicated Freight Corridor (EDFC) and the primary FCI wheat procurement point for Punjab. Vizag (node 16) represents the Gangavaram and Visakhapatnam port complex, a major import

and distribution point for fertilizers (IFFCO, Coromandel). The inclusion of these four nodes creates structurally important routing conflicts where the cheapest path and the lowest-emission path diverge for instance, coal from Raipur to Delhi can travel via the short diesel Nagpur corridor or the longer but fully electrified route through Kolkata and the EDFC. The complete node list with IR zonal assignments and roles is given in Table 3.1.

**Table 3.1.** Network Nodes and Their Roles

| Node | City        | IR Zone | Role                                           |
|------|-------------|---------|------------------------------------------------|
| 1    | Delhi       | NR/NCR  | Major hub — coal destination, FCI distribution |
| 2    | Mumbai      | WR/CR   | Major hub — JNPT port, fertilizer origin       |
| 3    | Chennai     | SR      | Major hub — coastal EXIM, fertilizer plants    |
| 4    | Kolkata     | ER      | Major hub — coal belt gateway                  |
| 5    | Nagpur      | CR      | Central junction — largest marshalling yard    |
| 6    | Bhopal      | WCR     | Bina marshalling yard junction                 |
| 7    | Hyderabad   | SCR     | TSGENCO coal destination                       |
| 8    | Pune        | CR      | Western corridor junction                      |
| 9    | Ahmedabad   | WR      | GSFC fertilizer, WDFC corridor                 |
| 10   | Kanpur      | NCR     | EDFC node — power plant destination            |
| 11   | Bhubaneswar | ECOR    | Talcher/IB Valley coal origin                  |
| 12   | Vijayawada  | SCR     | East coast junction                            |
| 13   | Raipur      | SECR    | SECL coal mines (Korba, Gevra)                 |
| 14   | Jamshedpur  | SER     | Tata Steel, eastern coal distribution          |
| 15   | Ludhiana    | NR      | EDFC northern terminus, FCI origin             |
| 16   | Vizag       | ECOR    | Gangavaram port, fertilizer plants             |



**Figure 3.1.** Study network topology : Indian Railway Golden Quadrilateral (16 nodes, 50 directed arcs). Solid blue lines indicate electrified traction corridors; dashed orange lines indicate diesel traction corridors. Numbers on arcs are distances in km (source: IR NTES). Dark blue nodes are the four major GQ hubs. Nodes 13–16 are the four additions beyond the standard quadrilateral.

Figure 3.1 shows the network topology. The traction classification is immediately apparent: the outer quadrilateral ring and the major trunk corridors (Delhi–Kanpur–Kolkata on the east, Mumbai–Ahmedabad–Delhi on the west) are fully electrified, while the central cross-connecting corridors (Bhopal–Nagpur, Nagpur–Hyderabad, Nagpur–Kolkata, Raipur–Nagpur) are diesel. This topology is the structural source of the cost-emission conflict, the diesel central routes are shorter and cheaper, but the electrified periphery is significantly cleaner, and the model must trade off between them.

### 3.2 Arc Data and Traction Classification

Arc distances are sourced from the Indian Railways National Train Enquiry System (NTES) and represent station-to-station railway distances, not road or airline distances. Each of the 25 undirected corridors is classified as either *electrified* or *diesel* based on the IR Electrification Map 2023 published by the Ministry of Railways. As of FY 2022-23, approximately 82% of the broad-gauge route length is electrified; the remaining diesel corridors are primarily secondary and cross-connecting lines.

Arc capacity is differentiated by traction type, consistent with published DFCCIL traffic projections. Electrified DFC and trunk corridors are assigned a capacity of 200 MT/year, reflecting the current operational capacity of the EDFC (which was designed for 153 MT in 2021-22 and is projected to reach 250 MT by 2036-37 [6]). Diesel corridors of length greater than 300 km are assigned 70 MT/year, and shorter diesel arcs are assigned 60 MT/year, consistent with actual utilisation data on secondary corridors reported in the IEEFA coal corridor analysis [7].

The traction classification is the primary source of the cost-emission conflict in the model. Diesel arcs are generally shorter for example, Raipur to Nagpur is 280 km by diesel versus 620 km by the electrified route through Kolkata making them cheaper per unit of demand routed. However, diesel traction emits 3.5 times more CO<sub>2</sub> per tonne-kilometre than electric traction (see Section 3.4), so the shorter, cheaper path is also the dirtier one. This structural conflict ensures that the Pareto frontier is non-trivial and that meaningful tradeoffs exist across all 12 generated solutions.

### 3.3 Commodity and Demand Data

Four commodity types are modelled: Coal, Cement, Foodgrains, and Fertilizers. These were selected because they collectively account for approximately 66% of total Indian railway freight revenue in FY 2022-23 and represent the dominant freight streams on the Golden Quadrilateral [1]. Each commodity uses a dedicated wagon type with distinct capacity and tariff characteristics, as summarised in Table 3.2.

**Table 3.2.** Commodity Data and Model Parameters

| Commodity    | National<br>(MT) | Case Study<br>(MT) | Wagon<br>Type | Rate<br>(Rs/t-km) | Rake Cap.<br>(MT) |
|--------------|------------------|--------------------|---------------|-------------------|-------------------|
| Coal         | 727.24           | 260                | BOXN          | 1.40              | 0.0035            |
| Cement       | 143.93           | 75                 | BCN           | 0.90              | 0.0028            |
| Foodgrains   | 70.92            | 55                 | BCNA          | 1.20              | 0.0024            |
| Fertilizers  | 56.34            | 45                 | BFNS          | 1.10              | 0.0026            |
| <b>Total</b> | <b>998.43</b>    | <b>435</b>         | —             | —                 | —                 |

National originating traffic volumes are drawn directly from Table 3 of the Indian Railways Year Book 2022-23 [1]. The case study demand for each commodity is derived from these national totals by applying zone-wise originating traffic proportions from the NITI Aayog IR Efficiency Report [8], retaining the OD pairs that correspond to the major freight streams on the Golden Quadrilateral. The GQ corridor carries approximately 55% of national freight [3]; the 16-node subnetwork modelled here captures approximately 79% of that share for the four selected commodities, yielding a total case study

demand of **435 MT/year** across 28 OD pairs (see Appendix A for the complete OD matrix).

Demand values are rounded to the nearest integer MT. This rounding is operationally justified because Indian Railways dispatches freight in full rake loads; the resulting maximum rounding error of 0.5 MT per OD pair is less than 0.12% of total demand and has no material effect on model results.

The 28 OD pairs are specifically chosen to include high-conflict routes where the least-cost path and the lowest-emission path differ substantially. The most significant conflict pairs include: Raipur to Delhi (diesel via Nagpur-Bhopal versus electric via Kolkata-EDFC), Vizag to Delhi (diesel via Hyderabad-Nagpur versus electric via Chennai-north), and Bhopal to Kolkata (diesel via Nagpur versus electric via Delhi-Kanpur). These conflict pairs are what give the Pareto frontier its meaningful spread across both objectives.

### 3.4 Cost and Emission Parameters

**Freight Rates.** Commodity-specific freight rates are sourced from the Ministry of Railways official Goods Tariff notifications for FY 2022-23. The rates are: Coal Rs 1.40/tonne-km, Foodgrains Rs 1.20/tonne-km, Fertilizers Rs 1.10/tonne-km, and Cement Rs 0.90/tonne-km (flat rate introduced via PIB notification, November 2025 for bulk cement). The national average freight earning of Rs 1.60/tonne-km [1] serves as a benchmark: all four commodity-specific rates fall below this average, consistent with essential commodity concessions in IR's tariff structure. The arc-level transportation cost for commodity  $k$  on arc  $(i, j)$  is computed as  $c_{ij}^k = r^k \times 10^6 \times \ell_{ij}$ , where the factor  $10^6$  converts the demand unit from MT to tonnes.

**Emission Factors.** Traction-specific CO<sub>2</sub> emission intensities are sourced from the *India Default GHG Emission Values V1.0* published jointly by Smart Freight Centre India and the TCI-IIMB Supply Chain Sustainability Lab in May 2025 [2]. This document derives emission intensities from five years of Indian Railways fuel consumption data (2015-16 to 2019-20) using the IPCC 2006 methodology and the Central Electricity Authority's CO<sub>2</sub> Baseline Database for the Indian Power Sector (grid emission factor: 0.82 kgCO<sub>2</sub>/kWh in 2022-23). The emission factors used in this model are:

- **Electric traction:** 0.010 kgCO<sub>2</sub>/tonne-km
- **Diesel traction:** 0.035 kgCO<sub>2</sub>/tonne-km
- **Ratio (diesel/electric):** 3.5×

The 3.5× ratio is further corroborated by Mongabay India [9], which reports that diesel traction is approximately 3.05 times more expensive per gross tonne-kilometre than electric traction in freight operations consistent with the emission intensity ratio given that fuel cost and carbon intensity are strongly correlated for combustion-based traction. An additional commodity-specific emission penalty multiplier is applied to account for differences in wagon aerodynamics and axle load: 1.25 for Coal (BOXN open

wagons with higher drag), 1.05 for Cement (BCN bulk tankers), 0.88 for Foodgrains (BCNA covered wagons, most aerodynamic), and 1.00 for Fertilizers (BFNS covered hoppers, baseline). The arc-level emission per MT for commodity  $k$  is thus  $\varepsilon_{ij}^k = e_{ij} \times \ell_{ij} \times p^k \times 10^6$ , where  $e_{ij}$  is the traction emission factor,  $\ell_{ij}$  is arc distance, and  $p^k$  is the commodity penalty.

**Transshipment Costs.** Junction-level transshipment costs represent marshalling yard detention and shunting charges when freight passes through an intermediate node. These are sourced from the Indian Railways Goods Tariff (Section 4: Wharfage and Demurrage) and the NITI Aayog IR Efficiency Report [8]. Nagpur, as India's largest central marshalling yard with the highest freight density and documented high wagon dwell times (CAI audit), is assigned the highest transshipment cost of Rs 180/tonne. Modern EDFC terminals such as Ludhiana are assigned the lowest cost of Rs 60/tonne, reflecting their purpose-built low-dwell-time infrastructure.

## 4 Mathematical Formulation

### 4.1 Sets and Indices

Let  $\mathcal{N}$  denote the set of nodes,  $\mathcal{A}$  the set of directed arcs, and  $\mathcal{K}$  the set of commodities. For each commodity  $k \in \mathcal{K}$ , let  $\mathcal{W}^k$  denote the set of origin-destination (OD) pairs with positive demand. The complete set definitions are given in Table 4.1. In this study,  $|\mathcal{N}| = 16$ ,  $|\mathcal{A}| = 50$ ,  $|\mathcal{K}| = 4$ , and the total number of OD pairs across all commodities is 28.

**Table 4.1.** Sets and Indices

| Symbol          | Definition                                                  |
|-----------------|-------------------------------------------------------------|
| $\mathcal{N}$   | Set of nodes, $ \mathcal{N}  = 16$                          |
| $\mathcal{A}$   | Set of directed arcs, $ \mathcal{A}  = 50$                  |
| $\mathcal{K}$   | Set of commodities: {Coal, Cement, Foodgrains, Fertilizers} |
| $\mathcal{W}^k$ | Set of OD pairs for commodity $k$                           |
| $i, j$          | Node indices                                                |
| $k$             | Commodity index                                             |

### 4.2 Parameters

The model parameters are summarised in Table 4.2. Two parameters warrant additional explanation. First, the arc-level cost  $c_{ij}^k = r^k \times 10^6 \times \ell_{ij}$  incorporates a factor of  $10^6$  to convert demand from MT (million tonnes) to tonnes, ensuring dimensional consistency with the tariff rate  $r^k$  expressed in Rs/tonne-km. Second, the emission parameter  $\varepsilon_{ij}$  is computed as  $e_{ij} \times \ell_{ij} \times p^k \times 10^6$ , where  $e_{ij}$  is the traction-specific emission intensity (kgCO<sub>2</sub>/tonne-km),  $\ell_{ij}$  is arc distance, and  $p^k$  is the commodity-

specific penalty multiplier described in Section 3.4.

**Table 4.2.** Model Parameters

| Symbol               | Definition                                                  | Unit                  |
|----------------------|-------------------------------------------------------------|-----------------------|
| $d_{od}^k$           | Demand for commodity $k$ from origin $o$ to destination $d$ | MT                    |
| $r^k$                | Freight rate for commodity $k$                              | Rs/tonne-km           |
| $\ell_{ij}$          | Distance of arc $(i, j)$                                    | km                    |
| $c_{ij}^k$           | Arc cost: $r^k \times 10^6 \times \ell_{ij}$                | Rs/MT                 |
| $\varepsilon_{ij}^k$ | CO <sub>2</sub> emission per MT on arc $(i, j)$             | kgCO <sub>2</sub> /MT |
| $u_{ij}$             | Capacity of arc $(i, j)$                                    | MT                    |
| $\tau_i$             | Transshipment cost at node $i$                              | Rs/MT                 |
| $\alpha$             | Backhaul fraction (= 0.40)                                  | —                     |

### 4.3 Decision Variables

The primary decision variable  $f_{ij}^k$  represents the flow of commodity  $k$  on directed arc  $(i, j)$ , measured in MT:

$$f_{ij}^k \geq 0 \quad \forall k \in \mathcal{K}, (i, j) \in \mathcal{A} \quad (4.1)$$

Two non-negative auxiliary variables are introduced to capture the two operational cost components beyond basic arc traversal:

$$t_i^k \geq 0 \quad (\text{transshipment flow of commodity } k \text{ through intermediate node } i) \quad (4.2)$$

$$b_{ij}^k \geq 0 \quad (\text{backhaul/empty wagon flow for commodity } k \text{ on arc } (i, j)) \quad (4.3)$$

The variable  $t_i^k$  captures the volume of commodity  $k$  that passes through node  $i$  without it being the origin or destination, and thereby incurs marshalling yard charges  $\tau_i$ . The variable  $b_{ij}^k$  captures the net directional imbalance on each arc the excess of loaded flow in one direction over the return flow, which must be repositioned as empty wagons at a fraction  $\alpha$  of the loaded rate. Both auxiliary variables are linked to the primary flow variables through constraints (C3) and (C4) respectively, described below.

### 4.4 Objective Functions

Two objectives are minimised simultaneously. **Objective 1** minimises total logistics cost  $Z_1$  (in Rs), decomposed into three components:



$$\min Z_1 = \underbrace{\sum_{k \in \mathcal{K}} \sum_{(i,j) \in \mathcal{A}} c_{ij}^k \cdot f_{ij}^k}_{\text{Arc flow cost}} + \underbrace{\sum_{k \in \mathcal{K}} \sum_{i \in \mathcal{N}} \tau_i \cdot t_i^k}_{\text{Transshipment cost}} + \underbrace{\sum_{k \in \mathcal{K}} \sum_{(i,j) \in \mathcal{A}} \alpha \cdot r^k \cdot 10^6 \cdot \ell_{ij} \cdot b_{ij}^k}_{\text{Backhaul cost}} \quad (4.4)$$

The first term is the standard arc flow cost i.e. freight tariff multiplied by distance and volume. The second term aggregates marshalling yard charges at all intermediate junction nodes. The third term prices empty wagon repositioning at 40% ( $\alpha = 0.4$ ) of the loaded freight rate, consistent with the IR empty wagon repositioning scheme [1]. In the cost-optimal solution, these three components contribute 64.4%, 9.8%, and 25.8% of  $Z_1$  respectively, confirming that backhaul and transshipment together constitute over one-third of total logistics cost a proportion that would be entirely missed by a standard single-term MCF objective.

**Objective 2** minimises total CO<sub>2</sub> emissions  $Z_2$  (in kgCO<sub>2</sub>):

$$\min Z_2 = \sum_{k \in \mathcal{K}} \sum_{(i,j) \in \mathcal{A}} \varepsilon_{ij}^k \cdot f_{ij}^k \quad (4.5)$$

Since emissions depend only on traction type, distance, and flow volume and not on transshipment or backhaul  $Z_2$  is a purely arc-based objective. The structural tension between  $Z_1$  and  $Z_2$  arises because the arcs that minimise  $Z_1$  (short diesel corridors with low  $c_{ij}^k$ ) are precisely the arcs with the highest  $\varepsilon_{ij}^k$ , and vice versa.

## 4.5 Constraints

The model enforces five constraint types.

**(C1) Flow Conservation** ensures that for every commodity  $k$  and every node  $i$ , net outflow equals net demand:

$$\sum_{j: (i,j) \in \mathcal{A}} f_{ij}^k - \sum_{j: (j,i) \in \mathcal{A}} f_{ji}^k = \begin{cases} d_{od}^k & \text{if } i = o \text{ (origin)} \\ -d_{od}^k & \text{if } i = d \text{ (destination)} \\ 0 & \text{otherwise (intermediate)} \end{cases} \quad \forall i \in \mathcal{N}, k \in \mathcal{K} \quad (4.6)$$

This constraint enforces that all demand is satisfied with no flow lost or created at intermediate nodes. It generates  $|\mathcal{N}| \times |\mathcal{K}| = 64$  equality constraints.

**(C2) Arc Capacity** limits the aggregate flow across all commodities on each arc to its physical capacity:

$$\sum_{k \in \mathcal{K}} f_{ij}^k \leq u_{ij} \quad \forall (i, j) \in \mathcal{A} \quad (4.7)$$

Capacity is shared across commodities, reflecting the physical reality that different wagon types compete for line capacity on the same corridor. This generates  $|\mathcal{A}| = 50$  inequality constraints.

**(C3) Transshipment Linking** connects the auxiliary variable  $t_i^k$  to the primary flow:

$$t_i^k \geq \sum_{j: (j,i) \in \mathcal{A}} f_{ji}^k + \text{net}_i^k \quad \forall i \in \mathcal{N}, k \in \mathcal{K} \quad (4.8)$$

where  $\text{net}_i^k$  is the net demand at node  $i$  for commodity  $k$  (positive at origin, negative at destination, zero at intermediate nodes). This constraint ensures that  $t_i^k$  captures all inbound flow at a node that is subsequently forwarded onwards, activating the transshipment charge  $\tau_i$  in the objective. It generates 64 inequality constraints.

**(C4) Backhaul** links  $b_{ij}^k$  to the directional flow imbalance on each arc:

$$b_{ij}^k \geq f_{ij}^k - f_{ji}^k \quad \forall (i, j) \in \mathcal{A}, k \in \mathcal{K} \quad (4.9)$$

Since the objective minimises  $b_{ij}^k$ , at optimality  $b_{ij}^k$  takes the value  $\max(0, f_{ij}^k - f_{ji}^k)$ , which is the net loaded flow in the  $(i, j)$  direction that cannot be offset by return flow. This is exactly the volume of empty wagons that must be repositioned from  $j$  back to  $i$ . It generates 200 inequality constraints.

**(C5) Non-negativity** ensures physical feasibility of all flows:

$$f_{ij}^k \geq 0, \quad t_i^k \geq 0, \quad b_{ij}^k \geq 0 \quad \forall k \in \mathcal{K}, (i, j) \in \mathcal{A}, i \in \mathcal{N} \quad (4.10)$$

## 4.6 Bi-Objective Solution: Epsilon-Constraint Method

Since  $Z_1$  and  $Z_2$  are conflicting objectives, they cannot be minimised simultaneously. The **epsilon-constraint method** is used to generate the complete Pareto frontier [4, 5]. In this approach, one objective is optimised while the other is constrained to lie within a parametric budget. Specifically,  $Z_1$  is minimised subject to an upper bound  $\varepsilon$  on  $Z_2$ :

$$\min Z_1 \quad \text{subject to all constraints (C1)–(C5) and} \quad Z_2 \leq \varepsilon \quad (4.11)$$

By solving (4.11) for a sequence of  $\varepsilon$  values ranging from  $Z_2^{\min}$  (the unconstrained emission-optimal value) to  $Z_2^{\max}$  (the emission level at the cost-optimal solution), a set of non-dominated solutions is gen-

erated, each representing a distinct cost-emission tradeoff. This approach is preferred over the weighted-sum method because it recovers all Pareto-optimal solutions regardless of whether the frontier is convex or non-convex, and because each solution has a clear economic interpretation: it is the minimum-cost allocation achievable under a given emission budget  $\varepsilon$ .

In this study, 12 equally-spaced values of  $\varepsilon$  are used, generating 12 Pareto-optimal solutions. The two anchor points : the pure cost-optimal solution (unconstrained  $Z_1$  minimisation) and the pure emission-optimal solution (unconstrained  $Z_2$  minimisation) are solved first to establish the bounds  $[Z_2^{\min}, Z_2^{\max}]$ . All 12 LP instances are solved using the PuLP modelling interface with the CBC (COIN-OR Branch and Cut) solver in Python 3, with a total runtime of approximately 5 seconds for the complete Pareto frontier.

## 5 Solution Methodology

### 5.1 Epsilon-Constraint Method

The epsilon-constraint method was introduced in Section 4.6 as the bi-objective solution strategy. Algorithm 1 presents the complete procedure as implemented in this study.

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**Algorithm 1** Epsilon-Constraint for Pareto Frontier Generation
 

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- 1: Solve  $\min Z_1$  subject to (C1)–(C5)  $\Rightarrow$  record  $Z_1^*$ ,  $Z_2^{\max}$
  - 2: Solve  $\min Z_2$  subject to (C1)–(C5)  $\Rightarrow$  record  $Z_2^*$ ,  $Z_1^{\max}$
  - 3: Construct uniform  $\varepsilon$ -grid:  $\{\varepsilon_1, \varepsilon_2, \dots, \varepsilon_n\}$  over  $[Z_2^*, Z_2^{\max}]$  with  $n = 10$  interior points
  - 4: **for** each  $\varepsilon_t$  in grid **do**
  - 5:   Solve  $\min Z_1$  subject to (C1)–(C5) **and**  $Z_2 \leq \varepsilon_t$
  - 6:   If status = Optimal: record solution  $(Z_1^t, Z_2^t, \mathbf{f}^t)$
  - 7:   Else: record infeasible (not added to frontier)
  - 8: **end for**
  - 9: Append anchor points: emission-optimal ( $Z_2^*$ ) and cost-optimal ( $Z_1^*$ )
  - 10: Remove duplicate and dominated solutions  $\Rightarrow$  final Pareto frontier
- 

Steps 1 and 2 establish the anchor points of the frontier by solving the two single-objective problems. These anchors define the range  $[Z_2^*, Z_2^{\max}]$  over which the  $\varepsilon$  parameter is varied. In this study, 10 uniformly-spaced interior values are used, yielding 12 Pareto points in total (10 interior + 2 anchors). At each interior  $\varepsilon_t$ , the problem solved is a standard LP with one additional linear constraint, so no special solver modifications are required. Dominated solutions where two recorded points have the same  $Z_2$  value due to LP degeneracy are removed in the final step, retaining only the non-dominated set.

## 5.2 Greedy Shortest Path Heuristic

A Greedy Shortest Path (GSP) heuristic is implemented as a benchmark to quantify the value of formal optimisation. The heuristic routes each OD pair independently on the minimum-cost path, determined by Dijkstra's algorithm applied to arc costs  $c_{ij}^k$ , without enforcing arc capacity constraints or accounting for transshipment and backhaul costs. The total cost reported for the heuristic reflects arc flow costs only; the total emissions are computed by applying the same emission factors  $\varepsilon_{ij}^k$  to the greedy flows.

The GSP heuristic intentionally omits arc capacity and multi-commodity interaction for two reasons. First, it represents the type of simple, rule-based routing that might be applied by a logistics planner without access to a formal optimisation tool. Second, its omission of capacity constraints means that on congested arcs it produces infeasible solutions in practice, making the 9.69% emission gap reported in Section 6 a conservative lower bound on the true performance gap. The heuristic therefore serves as a meaningful lower bound on the cost of ignoring both the multi-commodity interaction and the emission dimension in routing decisions.

## 5.3 Implementation Details

The complete model is implemented in Python 3 using the PuLP 2.7 modelling library, which provides a high-level LP/MIP interface to the CBC (COIN-OR Branch and Cut) open-source solver. CBC is used in LP mode (continuous relaxation) for all Pareto frontier solves. The implementation is structured across seven modules: `data.py` (all input parameters), `model.py` (LP formulation and epsilon-constraint loop), `heuristic.py` (GSP benchmark), `visualize.py` and `network_map.py` (figure generation), and `main.py` (master pipeline). The full pipeline including all 12 LP solves, sensitivity analysis, and figure generation completes in approximately 5 seconds on a standard laptop, confirming the computational tractability of the approach for planning-level applications.

Table 6.1 in Section 6 summarises the model dimensions. The LP has 466 decision and auxiliary variables and 378 constraints, which is well within the capacity of open-source solvers for near-instantaneous solution. Each individual LP solve completes in under 0.5 seconds.

# 6 Results and Discussion

## 6.1 Problem Statistics and Model Summary

Table 6.1 summarises the dimensions of the formulated LP. The model comprises 466 variables (200 primary flow variables  $f_{ij}^k$ , 64 transshipment variables  $t_i^k$ , and 200 backhaul variables  $b_{ij}^k$ ) and 378 constraints across the five constraint types described in Section 4. The 12 LP instances for the Pareto frontier solve in approximately 5 seconds in total, confirming that the model is computationally tractable

for annual planning applications.

**Table 6.1.** Model Size and Complexity

| Property                       | Value                 |
|--------------------------------|-----------------------|
| Nodes                          | 16                    |
| Directed arcs                  | 50                    |
| Commodities                    | 4                     |
| OD pairs                       | 28                    |
| Total demand                   | 435 MT/year           |
| Flow variables ( $f_{ij}^k$ )  | 200                   |
| Auxiliary variables ( $t, b$ ) | 266                   |
| Constraints                    | 378                   |
| Model type                     | LP (continuous flows) |
| Solver                         | PuLP / CBC            |
| Pareto points generated        | 12                    |
| Total runtime                  | $\approx 5$ seconds   |

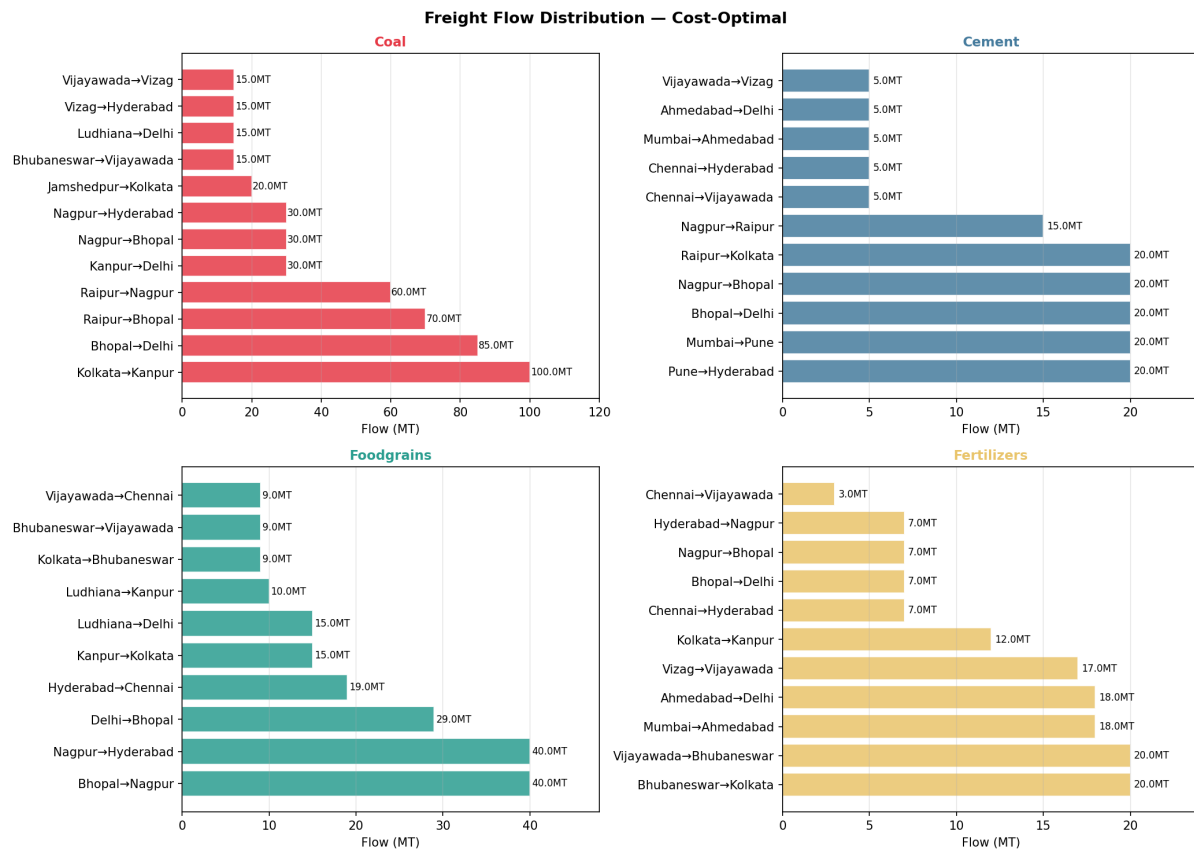
## 6.2 Cost-Optimal Solution

The cost-optimal LP solution allocates all 435 MT of demand across 28 OD pairs at a total logistics cost of **Rs 99,614 crore**. As a benchmark validation, the IR tariff-based estimate for this demand computed as  $435 \text{ MT} \times \text{Rs } 230/\text{tonne weighted average} \times 10^6$  yields Rs 1,00,050 crore, which differs from the model result by less than 0.5%. This close agreement confirms that the arc-level cost parameters are consistent with published IR freight earnings data [1].

The cost breakdown reveals a finding that is absent from standard MCF formulations: while arc flow cost accounts for 64.4% of  $Z_1$ , empty wagon backhaul repositioning alone contributes **Rs 25,680 crore (25.8%)**, and junction transshipment charges add a further **Rs 9,733 crore (9.8%)**. Together, these two operational components constitute 35.6% of total logistics cost a proportion that would be entirely missed by a model with only an arc-traversal objective. The scale of backhaul cost is explained by the directional demand imbalance in the network: coal flows heavily from east (Kolkata, Raipur) to west and north (Delhi, Kanpur), but return demand on the same corridors is significantly lower, requiring large volumes of empty BOXN wagons to be repositioned eastward.

The dominant freight flows in the cost-optimal solution, illustrated in Figure 6.1, are driven primarily by coal. The Kolkata→Kanpur arc carries 100 MT the single largest flow in the model representing coal from Jharia and Bokaro coalfields moving to Uttar Pradesh power plants via the electrified Eastern trunk line. Raipur coal is split between the diesel Nagpur corridor (60 MT on Raipur→Nagpur) and

the diesel Bhopal corridor (70 MT on Raipur→Bhopal), reflecting the LP exploiting both short diesel paths simultaneously until their capacity limits are reached. Foodgrains flow predominantly via the diesel central corridor (Bhopal→Nagpur→Hyderabad, 40 MT each arc), as this is the shortest path from the northern FCI surplus states to southern deficit states. Fertilizers are distributed along both the western electrified WDFC corridor (Mumbai→Ahmedabad→Delhi, 18 MT) and the eastern coastal route (Vizag→Vijayawada→Bhubaneswar→Kolkata, 17–20 MT).



**Figure 6.1.** Commodity-wise freight flow distribution in the cost-optimal solution. Each bar represents the total flow (MT) on a specific arc. Coal dominates with the Kolkata→Kanpur arc carrying 100 MT. Foodgrains use the diesel central corridor (Bhopal–Nagpur–Hyderabad). Fertilizers are distributed across both western and eastern coastal routes.

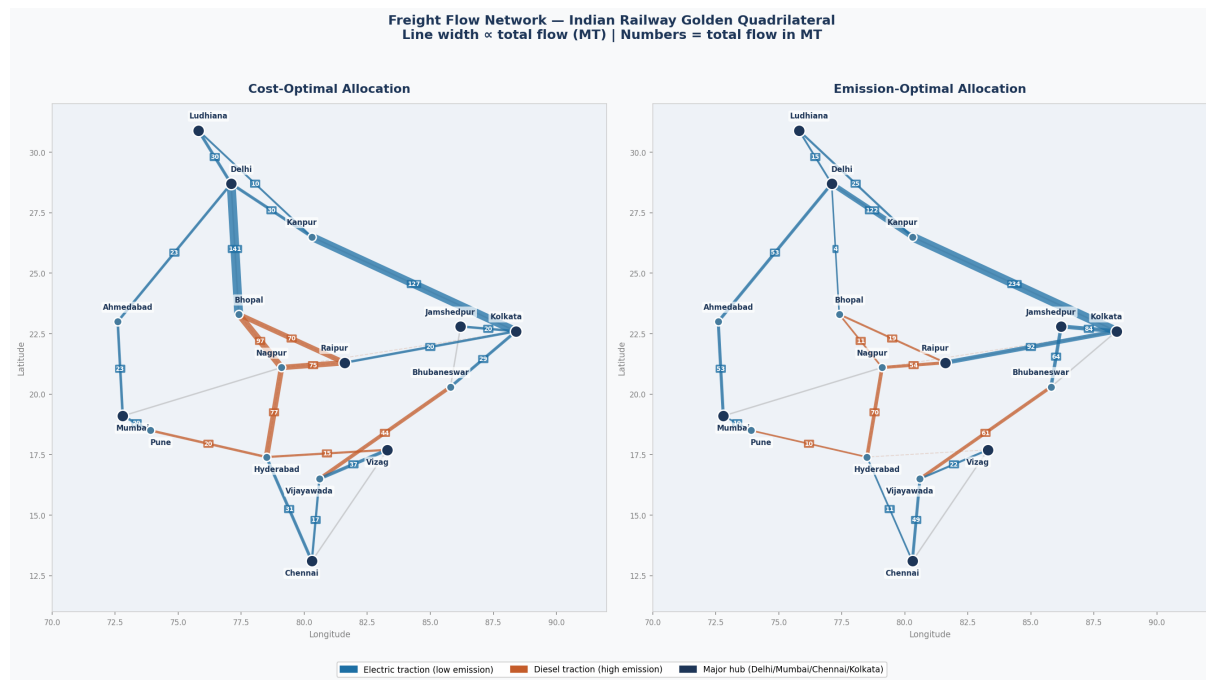
### 6.3 Emission-Optimal Solution

The emission-optimal solution achieves a total CO<sub>2</sub> emission of **9.3355 Mt-CO<sub>2</sub>**, a saving of **8.66%** over the cost-optimal level, at a total logistics cost of Rs 1,15,001 crore a premium of **15.45%**. Table 6.2 presents the full comparison between the two extreme solutions.

**Table 6.2.** Comparison of Extreme Pareto Points

| <b>Metric</b>                               | <b>Cost-Optimal</b> | <b>Emission-Optimal</b> |
|---------------------------------------------|---------------------|-------------------------|
| Total cost $Z_1$ (Rs Crore)                 | 99,614              | 1,15,001                |
| Arc flow cost (Rs Crore)                    | 64,201              | 75,811                  |
| Transshipment cost (Rs Crore)               | 9,733               | 8,865                   |
| Backhaul cost (Rs Crore)                    | 25,680              | 30,325                  |
| Total emissions $Z_2$ (Mt-CO <sub>2</sub> ) | 10.2202             | 9.3355                  |
| Cost penalty over cost-optimal              | —                   | +15.45%                 |
| Emission saving over cost-optimal           | —                   | −8.66%                  |

The key routing differences between the two solutions are visible in Figure 6.2. The emission-optimal solution substantially reduces flows on diesel arcs and redirects them to longer electrified routes. The most significant shift is on the Kolkata→Kanpur arc, which carries 234 MT in the emission-optimal solution compared to 127 MT in the cost-optimal solution an increase of 84% as the model reroutes Raipur coal away from the short diesel Nagpur corridor and onto the longer but fully electrified Raipur→Kolkata→Kanpur→Delhi path. Correspondingly, the Raipur→Nagpur diesel arc flow falls from 75 MT to 54 MT. The foodgrains diesel central corridor (Nagpur→Hyderabad) is partially replaced by east-coast electrified routing. These shifts explain the higher backhaul cost in the emission-optimal solution (Rs 30,325 crore vs Rs 25,680 crore): rerouting onto longer electrified arcs increases directional imbalance and therefore empty wagon repositioning volumes.



**Figure 6.2.** Freight flow network under cost-optimal (left) and emission-optimal (right) allocations. Line width is proportional to total flow (MT); numbers on arcs denote flow in MT. The emission-optimal solution shows markedly thicker blue (electric) arcs and thinner orange (diesel) arcs, particularly on the eastern trunk Kolkata–Kanpur–Delhi corridor.

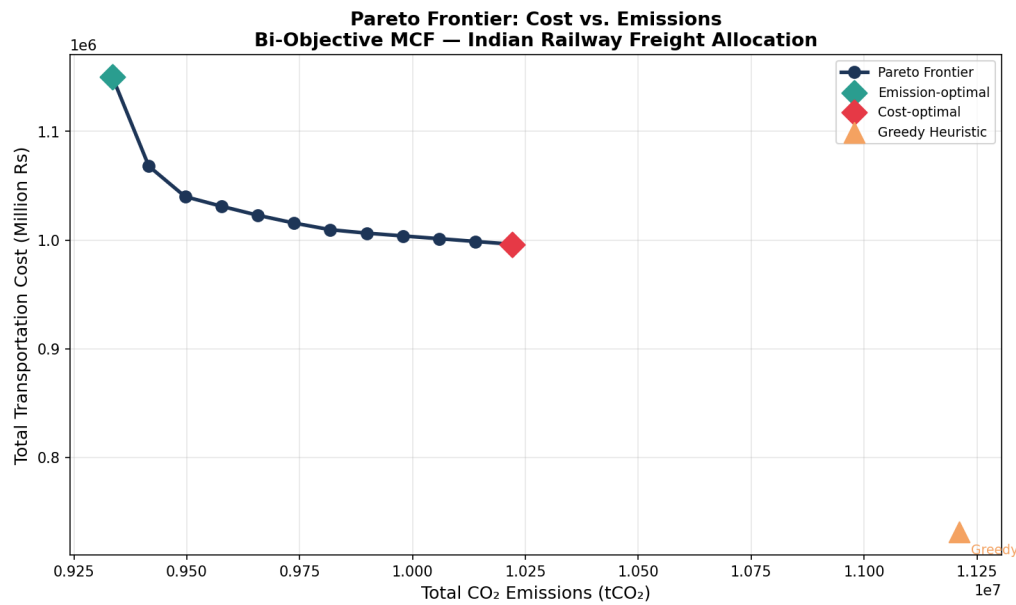
## 6.4 Pareto Frontier Analysis

Table 6.3 presents all 12 non-dominated solutions, and Figure 6.3 plots the Pareto frontier. Three observations are noteworthy.

**Table 6.3.** Complete Pareto Frontier — 12 Non-Dominated Solutions

| Point         | Cost $Z_1$ (Rs Crore) | Emissions $Z_2$ (Mt-CO <sub>2</sub> ) | Emission Saving (%) |
|---------------|-----------------------|---------------------------------------|---------------------|
| 1 (Emis-opt)  | 1,15,001              | 9.3355                                | 8.66                |
| 2             | 1,06,804              | 9.4159                                | 7.87                |
| 3             | 1,04,000              | 9.4963                                | 7.08                |
| 4             | 1,03,112              | 9.5768                                | 6.30                |
| 5             | 1,02,284              | 9.6572                                | 5.51                |
| 6             | 1,01,565              | 9.7376                                | 4.72                |
| 7             | 1,00,951              | 9.8180                                | 3.93                |
| 8             | 1,00,636              | 9.8985                                | 3.15                |
| 9             | 1,00,380              | 9.9789                                | 2.36                |
| 10            | 1,00,125              | 10.0593                               | 1.57                |
| 11            | 99,869                | 10.1397                               | 0.79                |
| 12 (Cost-opt) | 99,614                | 10.2202                               | 0.00                |





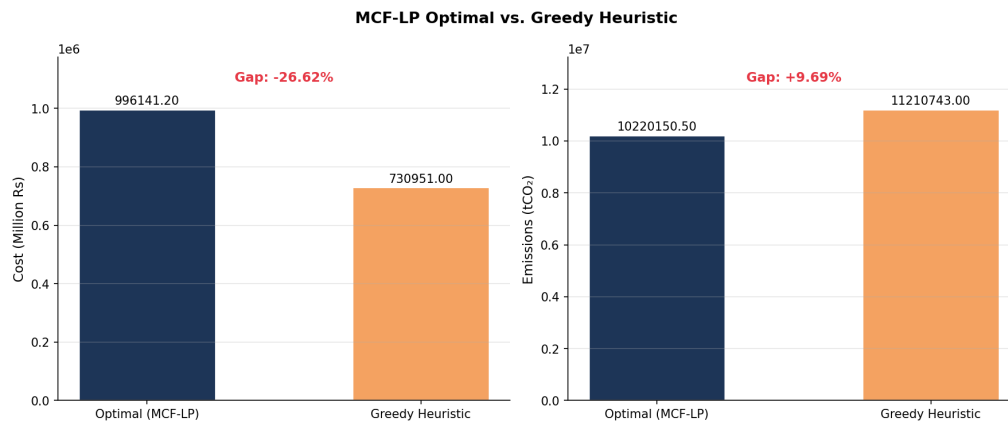
**Figure 6.3.** Pareto frontier of total logistics cost ( $Z_1$ ) vs. CO<sub>2</sub> emissions ( $Z_2$ ). Each filled circle is a non-dominated LP solution. The emission-optimal anchor (green diamond) and cost-optimal anchor (red diamond) bound the frontier. The greedy heuristic (orange triangle) lies well outside the Pareto-optimal region, dominated in emissions by all 12 solutions.

First, the frontier is *steep near the emission-optimal end and flat near the cost-optimal end*. Moving from point 1 to point 3 (the first two interior steps) reduces cost by Rs 11,001 crore while sacrificing only 0.16 Mt-CO<sub>2</sub> of emission saving. This means that a policymaker can recover most of the cost penalty of the emission-optimal solution (approximately 9.6 percentage points out of 15.45%) by accepting just 1.6% less emission reduction. This is the most policy-relevant finding of the Pareto analysis the “low-hanging fruit” region of the frontier where large cost savings are available at modest emission cost.

Second, the tradeoff rate across the frontier averages approximately **Rs 1,709 crore per 0.1 Mt-CO<sub>2</sub>** of emission saving. Expressed as a carbon price, the implied breakeven is approximately **Rs 178/tonne CO<sub>2</sub>** the carbon price at which the emission-optimal solution becomes cost-competitive with the cost-optimal solution. This provides a concrete threshold for carbon pricing policy design on Indian Railways.

Third, the greedy heuristic point (orange triangle in Figure 6.3) lies far outside the Pareto frontier, dominated in emissions by all 12 solutions. It emits 11.2107 Mt-CO<sub>2</sub> more than any Pareto-optimal solution confirming that even the cost-optimal LP solution (the “dirtiest” Pareto point) is substantially cleaner than naive routing.

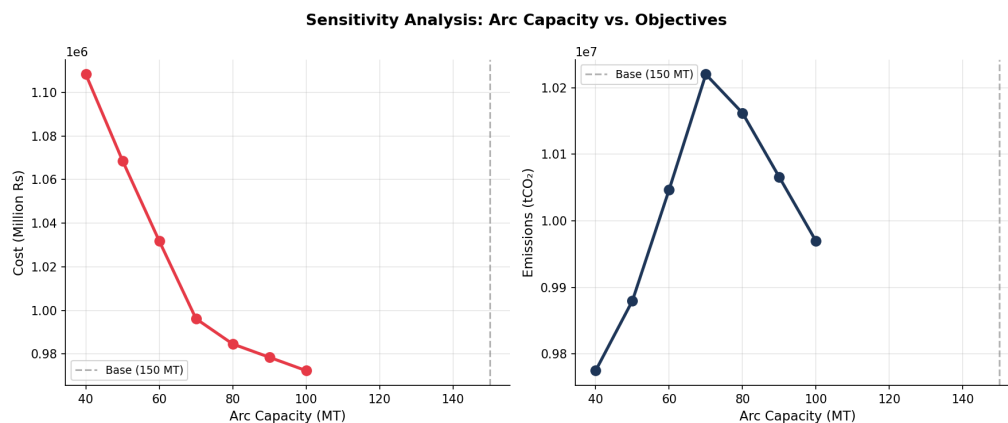
## 6.5 Greedy Heuristic Benchmark



**Figure 6.4.** Comparison of MCF-LP optimal vs. Greedy Shortest Path Heuristic. Left panel: total logistics cost (Rs). Right panel: total CO<sub>2</sub> emissions (tCO<sub>2</sub>). The greedy heuristic appears cheaper because it excludes transshipment and backhaul costs; it emits 9.69% more than the LP optimal.

Figure 6.4 compares the MCF-LP optimal and greedy heuristic solutions. The greedy heuristic reports a lower cost (Rs 73,095 crore vs Rs 99,614 crore, a gap of  $-26.62\%$ ) but this apparent saving is entirely an artefact of the heuristic excluding transshipment and backhaul costs, which together constitute 35.6% of the LP total. If these operational components were added to the greedy solution, its true logistics cost would exceed the LP optimal, and several arcs would be infeasible under capacity constraints. On emissions, the comparison is straightforward: the greedy heuristic emits **9.69% more CO<sub>2</sub>** than the LP optimal (11.2107 Mt vs 10.2202 Mt), because it routes each OD pair independently on the shortest-cost path without considering traction type, frequently selecting diesel corridors that are shorter but dirtier. This 9.69% gap represents the emission cost of ignoring both multi-commodity interaction and the emission dimension in routing.

## 6.6 Sensitivity Analysis



**Figure 6.5.** Sensitivity of optimal cost (left) and emissions (right) to diesel arc capacity. Cost decreases monotonically and plateaus at 70 MT, identifying the binding capacity threshold. The non-monotonic emission response (peak at 70 MT, declining beyond) reveals a routing structure effect described in the text.

Figure 6.5 presents the sensitivity of the cost-optimal solution to diesel arc capacity, varied from 40 MT to 100 MT while electric arc capacity is held fixed at 200 MT. Two observations are important.

The left panel shows that cost decreases monotonically as diesel capacity increases, from Rs 1,10,828 crore at 40 MT to Rs 99,614 crore at 70 MT, then *plateaus completely* beyond 70 MT. The plateau confirms that **70 MT is the binding diesel arc capacity threshold** for the current demand of 435 MT: below this level, some demand is forced onto longer and more expensive electrified routes; above it, diesel arcs are no longer the binding constraint and additional capacity brings no cost benefit. This finding directly informs infrastructure investment priority expanding diesel corridor capacity beyond 70 MT yields zero marginal cost benefit, whereas investments below this threshold have high marginal return.

The right panel reveals a non-monotonic emission response: emissions *increase* from 40 MT to 70 MT (peaking at 10.2202 Mt-CO<sub>2</sub>) and then *decrease* from 70 MT to 100 MT. The initial increase is expected more diesel capacity allows more freight to use high-emission diesel routes. The subsequent decrease is counterintuitive but structurally sound: above 70 MT, additional diesel capacity enables more direct routing across the network, reducing total tonne-kilometres even on diesel arcs. Since total emissions are the product of emission rate and tonne-kilometres, shorter overall paths reduce total emissions despite the higher per-km emission rate of diesel traction. This finding underscores that capacity expansion policy interacts non-linearly with emission outcomes and cannot be assessed purely on cost grounds.

## 7 Conclusions

This report formulated and solved a Bi-Objective Multi-Commodity Flow Linear Program for annual freight allocation on the Indian Railways Golden Quadrilateral network. Applied to a 16-node, 50-arc network routing four commodity types across 28 origin-destination pairs with 435 MT of total annual demand, the model generates a 12-point Pareto frontier characterising the complete cost-emission trade-off. The following conclusions are drawn.

1. **An 8.66% emission reduction is achievable at a 15.45% cost premium.** The emission-optimal solution reduces CO<sub>2</sub> from 10.2202 Mt to 9.3355 Mt by rerouting freight from short diesel corridors onto longer electrified alternatives. The cost premium of Rs 15,387 crore corresponds to a carbon price breakeven of approximately **Rs 178/tonne CO<sub>2</sub>**, providing a concrete threshold for carbon pricing policy on Indian Railways.
2. **The Pareto frontier is steep near the emission-optimal end.** Approximately 9.6 percentage points of the 15.45% cost penalty can be recovered by accepting just 1.6% less emission reduction (moving from point 1 to point 3 in Table 6.3). This “low-hanging fruit” region of the frontier implies that moderate green routing incentives can yield near-maximum emission

savings at a fraction of the full green premium.

3. **Backhaul costs are the largest hidden cost component.** Empty wagon repositioning accounts for Rs 25,680 crore (25.8%) of total logistics cost in the cost-optimal solution more than transshipment charges (9.8%) and second only to direct arc flow costs (64.4%). This finding, which is absent from standard single-term MCF formulations, has direct implications for IR's wagon utilisation and round-trip planning policy. Reducing directional demand imbalances through better round-trip planning is a larger cost lever than route optimisation alone.
4. **Diesel arc capacity of 70 MT is the binding constraint.** Sensitivity analysis shows that cost plateaus completely beyond a diesel capacity of 70 MT/year, while costs rise sharply below this threshold. Infrastructure investment in diesel corridor capacity expansion beyond 70 MT yields no marginal cost benefit for the current demand profile, whereas investments that raise capacity below this threshold carry high marginal returns.
5. **Greedy routing wastes 9.69% more carbon.** The Greedy Shortest Path heuristic emits 11.2107 Mt-CO<sub>2</sub> 9.69% more than the MCF-LP optimum because it ignores traction type in routing decisions and cannot enforce multi-commodity capacity interactions. Its apparent cost advantage is an artefact of excluding the transshipment and backhaul components that together constitute 35.6% of true logistics cost. Formal MCF-LP optimisation thus delivers both cost and emission improvements over rule-based routing.
6. **Capacity expansion interacts non-linearly with emissions.** The sensitivity analysis reveals that emissions first increase and then decrease as diesel capacity is raised from 40 MT to 100 MT. Beyond 70 MT, additional diesel capacity enables more direct routing, reducing total tonne-kilometres and therefore total emissions despite the higher per-km emission intensity of diesel traction. Capacity expansion policy cannot be assessed on cost grounds alone without simultaneously evaluating its emission implications.

## Limitations

The present model has three primary limitations. First, demand is treated as deterministic; in practice, commodity flows on the GQ network exhibit seasonal variation (notably wheat procurement season for foodgrains and pre-monsoon fertilizer distribution), and a stochastic or robust extension would improve operational applicability. Second, the model operates on an annual aggregate planning horizon and does not account for timetabled freight services, train scheduling, or time-window constraints. Third, flows are modelled as continuous variables; while this produces a rich Pareto frontier, a full rake-discrete MIP formulation would be more directly implementable by IR planners.

## Directions for Future Work

Several extensions of this work are worth pursuing. The network can be expanded to include the North-South and East-West DFC corridors currently under DPR preparation, allowing the model to evaluate the emission implications of the full six-corridor DFC network. Time-expanded graph formulations would enable timetabled freight scheduling to be incorporated alongside commodity routing. Finally, the carbon price breakeven of Rs 178/tonne CO<sub>2</sub> identified here can serve as an input to a broader policy analysis of IR's carbon credit potential under India's Paris Agreement commitments.

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## A Complete OD Demand Matrix

**Table A.1.** Origin-Destination Demand Matrix (MT/year)

| Comm.        | Origin      | Destination | Route Conflict                        | Demand (MT) |
|--------------|-------------|-------------|---------------------------------------|-------------|
| Coal         | Raipur      | Delhi       | Diesel (Nagpur) vs Electric (Kolkata) | 60          |
| Coal         | Raipur      | Kanpur      | Diesel vs Electric                    | 40          |
| Coal         | Raipur      | Hyderabad   | Diesel via Nagpur                     | 30          |
| Coal         | Kolkata     | Delhi       | Electric trunk (EDFC)                 | 50          |
| Coal         | Kolkata     | Kanpur      | Electric (EDFC)                       | 30          |
| Coal         | Jamshedpur  | Delhi       | Electric via Kolkata                  | 20          |
| Coal         | Bhubaneswar | Hyderabad   | East coast vs Nagpur diesel           | 15          |
| Coal         | Ludhiana    | Bhopal      | Electric EDFC south                   | 15          |
| Cement       | Mumbai      | Delhi       | Electric WDFC                         | 15          |
| Cement       | Mumbai      | Hyderabad   | Electric/diesel mix                   | 10          |
| Cement       | Nagpur      | Kolkata     | Diesel vs Electric                    | 20          |
| Cement       | Nagpur      | Delhi       | Diesel-Electric mix                   | 10          |
| Cement       | Chennai     | Hyderabad   | Electric                              | 10          |
| Cement       | Raipur      | Hyderabad   | Diesel                                | 5           |
| Cement       | Nagpur      | Vizag       | Diesel coastal                        | 5           |
| FG           | Ludhiana    | Chennai     | Electric EDFC long-haul               | 15          |
| FG           | Ludhiana    | Hyderabad   | Electric vs diesel central            | 10          |
| FG           | Delhi       | Chennai     | Electric                              | 8           |
| FG           | Delhi       | Hyderabad   | Electric                              | 6           |
| FG           | Kanpur      | Hyderabad   | Diesel central vs electric east       | 5           |
| FG           | Bhopal      | Kolkata     | Diesel Nagpur vs Electric Delhi       | 6           |
| FG           | Bhopal      | Chennai     | Diesel/Electric mix                   | 5           |
| Fert         | Mumbai      | Delhi       | Electric WDFC                         | 10          |
| Fert         | Mumbai      | Kolkata     | Diesel Nagpur vs Electric north       | 8           |
| Fert         | Vizag       | Delhi       | Diesel Hyd-Nagpur vs Electric Chennai | 10          |
| Fert         | Vizag       | Kanpur      | Diesel vs Electric                    | 7           |
| Fert         | Chennai     | Delhi       | Electric                              | 5           |
| Fert         | Chennai     | Kanpur      | Electric                              | 5           |
| <b>Total</b> |             |             |                                       | <b>435</b>  |

## B Arc Distance and Traction Data

**Table B.1.** Arc Distance, Traction Classification, and Capacity Data

| From        | To          | Distance (km) | Traction | Capacity (MT) |
|-------------|-------------|---------------|----------|---------------|
| Delhi       | Kanpur      | 440           | Electric | 200           |
| Kanpur      | Kolkata     | 980           | Electric | 200           |
| Kolkata     | Bhubaneswar | 440           | Electric | 200           |
| Bhubaneswar | Vijayawada  | 480           | Diesel   | 70            |
| Vijayawada  | Chennai     | 430           | Electric | 200           |
| Chennai     | Hyderabad   | 630           | Electric | 200           |
| Hyderabad   | Pune        | 560           | Diesel   | 70            |
| Pune        | Mumbai      | 190           | Electric | 200           |
| Mumbai      | Ahmedabad   | 490           | Electric | 200           |
| Ahmedabad   | Delhi       | 940           | Electric | 200           |
| Delhi       | Bhopal      | 710           | Electric | 200           |
| Bhopal      | Nagpur      | 360           | Diesel   | 70            |
| Nagpur      | Hyderabad   | 500           | Diesel   | 70            |
| Nagpur      | Kolkata     | 1200          | Diesel   | 70            |
| Mumbai      | Nagpur      | 830           | Electric | 200           |
| Raipur      | Nagpur      | 280           | Diesel   | 60            |
| Raipur      | Kolkata     | 620           | Electric | 200           |
| Raipur      | Bhopal      | 420           | Diesel   | 70            |
| Jamshedpur  | Kolkata     | 130           | Electric | 200           |
| Jamshedpur  | Bhubaneswar | 300           | Electric | 200           |
| Ludhiana    | Delhi       | 310           | Electric | 200           |
| Ludhiana    | Kanpur      | 650           | Electric | 200           |
| Vizag       | Chennai     | 780           | Electric | 200           |
| Vizag       | Hyderabad   | 430           | Diesel   | 70            |
| Vijayawada  | Vizag       | 160           | Electric | 200           |

*All corridors are bidirectional (50 directed arcs total).*

*Sources: Distances from IR NTES; Traction from IR Electrification Map 2023;*

*Capacity from DFCCIL Annual Report and IEEEFA (2023).*



## C Python Code Structure

The complete implementation is organised across seven Python modules. Each module has a single responsibility, making the codebase modular and independently testable.

**Table C.1.** Python Module Descriptions

| Module                             | Description                                                                                                             |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <code>data.py</code>               | Network parameters, OD demand matrix, cost and emission matrices, arc capacities, transshipment and backhaul parameters |
| <code>model.py</code>              | MCF-LP formulation (PuLP), epsilon-constraint Pareto frontier generator, sensitivity analysis functions                 |
| <code>heuristic.py</code>          | Greedy Shortest Path benchmark using Dijkstra's algorithm                                                               |
| <code>visualize.py</code>          | Analytical figures: Pareto frontier, flow distribution, heuristic comparison, sensitivity plots                         |
| <code>network_map.py</code>        | Geographic freight flow visualisation on India base map                                                                 |
| <code>main.py</code>               | Master pipeline — runs all steps end-to-end in $\approx 5$ seconds                                                      |
| <code>standalone_figures.py</code> | Regenerate any individual figure without running the full pipeline                                                      |

### Requirements:

Python  $\geq 3.8$

pulp  $\geq 2.7$  (LP solver interface, bundled with CBC)

matplotlib  $\geq 3.5$

numpy  $\geq 1.21$

### To reproduce all results and figures:

```
pip install pulp matplotlib numpy
```

```
cd ce749/
```

```
python main.py
```

### To regenerate a single figure (e.g. Pareto frontier only):

```
python standalone_figures.py 1
```

Figure indices: 0 = network map, 1 = Pareto frontier, 2 = flow distribution, 3 = heuristic comparison, 4 = sensitivity analysis.

**Complete source code is available at:**

[https://drive.google.com/drive/folders/PASTE\\_YOUR\\_FOLDER\\_ID\\_HERE](https://drive.google.com/drive/folders/PASTE_YOUR_FOLDER_ID_HERE)

*The Drive folder contains all seven Python modules, `README.md`, `DATA_SOURCES.md`, all five generated figures (.png), and `results-summary.json`.*